

N20 DC 3V Metal Gear Motor

Description

The [N20 DC 3V Metal Gear Motor](#) is a compact, high-torque, low-RPM DC motor widely used in small-scale robotics, DIY electronics, and automated projects. The motor features an all-metal gearbox for durability and wear resistance, and it comes in a variety of gear ratios to provide different combinations of speed and torque.

Common specifications

The specifications can vary based on the manufacturer and the specific gear ratio, but here is a typical example of an N20 DC 3V metal gear motor, according to

- **Rated Voltage:** 3V DC
- **Voltage Range:** 1.5V–5V DC
- **Shaft Type:** D-type, single-sided
- **Shaft Diameter:** 3 mm
- **Motor Body Size:** 12 mm diameter
- **Weight:** Approximately 10g

Common variations

- **High Speed:** An N20 with a 10:1 gear ratio offers a higher RPM, suitable for applications needing faster movement.
- **High Torque:** An N20 with a 50:1 or 298:1 gear ratio provides more torque for heavier loads, though with a lower RPM.
- **With Encoder:** Some N20 motors include an integrated encoder, which provides feedback for precise control over position and speed in robotics and automation projects.